(3 cm,3 cm) Q (18)、

(19) V_1' V_2 3

R_1' R_2

$$R_1' = 3R_2$$

V_1' O x

Q

$$R_1' = 3\ \text{cm}$$

O_1' (0 cm,3 cm)

$$R_2 = 1\ \text{cm}$$

O_2 (0 cm,1 cm)

$$x^2 + (y-1)^2 = 1 \qquad (20)$$

Q P (x_P, y_P)

P V_2 PQ Q P

P O_2P PQ P $O_1'Q$

H O_2 PH F $\triangle O_2PF$ $\triangle PQH$

$$\frac{\overline{O_2F}}{\overline{PF}} = \frac{\overline{PH}}{\overline{HQ}} \qquad (21)$$

$$\frac{x_P}{(1-y_P)} = \frac{(3-y_P)}{(3-x_P)} \qquad (22)$$

$$x_P^2 + y_P^2 - 4y_P = 3x_P - 3$$

$$x_P^2 + (y_P-1)^2 - 2y_P - 1 = 3x_P - 3$$

P (x_P, y_P) (20)

$$-2y_P = 3x_P - 3$$

$$y_P = \frac{3}{2}(1 - x_P) \qquad (23)$$

(20) P y

$$y_P = 0.354\ \text{cm} \qquad (24)$$

MN P

$M'N'$ V_2 P

Q

a、b b c

a、d b、c

b c v_0

a v_0 b

a b M_1

$v_0/2$ c v_0 d c d

M_2 $v_0/2$

M_1 $v_0/2$

X'

b c